

NAVDEEP SINGH

Data Engineer II

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PERSONAL STATEMENT

Data Engineer II with ~3 years of industry experience building scalable analytics pipelines and production-grade data products at ZS Associates. I am seeking to advance my expertise through a Masters in Data Science / Analytics, driven by a passion for turning complex data into actionable insights. My background spans cloud data platforms (Databricks, AWS), SQL/Python engineering, CI/CD pipelines, and stakeholder collaboration across global Finance and Procurement domains. I also bring a unique dimension through hands-on cybersecurity research - recognized as a Top 15 Bug Bounty Researcher in India by NCIIPC (Government of India) - reflecting strong analytical thinking, attention to detail, and a curiosity-driven mindset that I believe are equally valuable in data science.

EDUCATION

Bachelor of Technology — Computer Science

Chandigarh Group of Colleges, Mohali, Punjab, India | 2019 – 2023 | CGPA: 8.05/10

- Relevant coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Computer Networks, Operating Systems, Software Engineering
- Built a CLI-based penetration testing automation tool to identify vulnerabilities in web applications
- Active contributor to cybersecurity research throughout undergraduate years, culminating in Top 15 Bug Bounty Researcher ranking by NCIIPC (Government of India)
- Developed strong programming fundamentals in Python and SQL, directly applied in professional data engineering roles post-graduation

PROFESSIONAL EXPERIENCE

Data Engineer II | ZS Associates

January 2026 – Present | Buenos Aires, Argentina

- Twice recipient of the ZS Impact Award; received multiple client and ZS leadership appreciation emails for technical contributions and project impact – project requirements and my contribution track led to a 3-month deployment to the Buenos Aires office for real-time client and technical collaboration on a critical engagement
- Leading data platform modernisation, architecting end-to-end solutions across Finance and Procurement domains with data quality frameworks and governance standards for enterprise-wide adoption
- Driving pipeline architecture decisions, optimising ETL workflows for performance and cost efficiency on Databricks and AWS
- Collaborating with Finance leadership and Analytics teams to translate business requirements into robust data products; mentoring junior engineers on best practices and code standards

Data Engineer I | ZS Associates

September 2023 – December 2025 | Gurugram, India

Finance Analytics Data Product — Cost of Goods Manufactured Standardisation

- Designed and delivered a production-grade data pipeline consolidating financial calculation logic across 7+ global manufacturing sites into a unified, scalable solution serving Finance stakeholders
- Built a reusable SQL transformation framework with PySpark for large-scale processing, implementing robust data quality checks, exception handling, and reconciliation processes
- Designed SQL-based transformation layers following dbt-style modular modelling, testing, and documentation principles to ensure trusted, maintainable analytics datasets
- Developed automated monitoring and alerting system providing real-time job status notifications to business users, ensuring reliable data delivery
- Migrated pipelines from Prophecy ETL to Databricks for continuous flow of data with zero downtime, maintaining full functional equivalence while improving maintainability
- Recognised by Finance leadership for transforming manual, error-prone processes into auditable, scalable data products

AI-Powered Procurement Analytics Pipeline

- Led a team of 2 engineers building an end-to-end analytics pipeline processing multilingual procurement data with automated translation, AI classification (GPT API), and category standardisation
- Implemented a data validation framework handling NULL values, data type conversions, and quality controls for large-scale datasets; built caching and rate-limiting strategies for API reliability
- Designed and implemented GitLab CI/CD pipelines automating Databricks job deployment with environment-specific configurations, improving release velocity and reducing deployment errors

Backend Software Engineer | Suptho

June 2023 – September 2023 | Remote

- Developed GraphQL queries and mutations in TypeScript for a React Native mobile application, building robust backend APIs for user-facing features
- Designed and implemented backend data layer logic, ensuring type-safe API interfaces between mobile frontend and backend services
- Collaborated with the frontend team to optimise API performance, reduce payload sizes, and improve application responsiveness

INTERNSHIP EXPERIENCE

Cybersecurity Summer Intern | Gurugram Police — Haryana State Cyber Crime Investigating Centre

April 2021 – July 2021 | Gurugram, India

- Collaborated with SafeHouse Technologies and the Society for Safe Gurugram on cybersecurity research projects under Gurugram Police
- Conducted research on malware forensics, data farming techniques, cybercrime law, and digital forensics on seized Google Drive data
- Gained hands-on experience with investigative procedures and legal frameworks for cybercrime in India

TECHNICAL SKILLS

Programming Languages: Python (Advanced), SQL (Advanced), PySpark, TypeScript

Cloud Data Platforms: Databricks, Delta Lake, Unity Catalog, AWS (S3, Lambda)

Analytics Engineering: dbt-style modular SQL, ETL/ELT Design, Data Quality & Validation, Performance Optimisation

Data Orchestration: Workflow orchestration (Airflow-like patterns), Prophecy ETL, Pipeline Automation

DevOps & Version Control: Git, GitLab, CI/CD Pipelines, Automated Deployment

Data Governance: Data Reconciliation, Metadata Management, Lineage Tracking, Automated Alerting

Cybersecurity: OWASP Top 10, Penetration Testing, Vulnerability Assessment, API Security, Burp Suite

Other Tools: SharePoint Integration, Excel/CSV Processing, GraphQL

ACHIEVEMENTS

National Recognition

- Ranked in the Top 15 Bug Bounty Researchers in India (Q3 2021) by the National Critical Information Infrastructure Protection Centre (NCIIPC), Government of India - recognised among 307+ participating researchers
- Received 51+ security acknowledgments and awards from the Government of India, international companies, and private organisations for identifying and responsibly disclosing critical vulnerabilities in production systems

Bug Bounty & Vulnerability Disclosure Awards

- Swiggy: Identified and disclosed a medium-severity vulnerability (sensitive data exposure); rewarded with gift vouchers worth INR 5,000
- HackerRank: Discovered a valid Open Redirect vulnerability; acknowledged with HackerRank swag and official recognition from the Security Engineering team
- Taplock Corp (Canada): Identified a qualifying vulnerability; rewarded with USD 100 by the Bug Bounty team
- BLS AG (Switzerland GoBugFree platform): Accepted reports including Content Spoofing / Text Injection (CHF 300) and No Rate Limit on Login Page leading to potential account takeover (CHF 450)
- BLS AG Android App: Identified hardcoded API key exposure affecting multiple Google APIs; accepted with CHF 220 payout

- Threema (Switzerland): Accepted rate-limiting vulnerability in contact support form; CHF 50 payout
- bugbountyhub.com: Medium-severity finding Memory corruption due to missing rate limit; accepted with CHF 560 payout
- fidenty.ch: Unauthenticated WordPress database upgrade endpoint; accepted with CHF 200 payout
- bpost (Belgium - Intigriti platform): Google API key leak disclosed; resolved as low severity
- Platform161 (Netherlands): CVE-2020-13379 vulnerability in Grafana dashboard; rewarded with EUR 40
- BugV Inc.: Bounty payment of USD 20 for accepted security report

Penetration Testing Certification

- Virtually Testing Foundation: Penetration Testing Internship (October - December 2021); curriculum included OWASP Top 10, web application pentesting labs, Burp Suite, vulnerability exploitation, CTF challenges, and professional pentest report writing

CERTIFICATIONS & TRAINING

- Penetration Testing Internship (10 weeks), Virtually Testing Foundation, December 2021
- Cybersecurity Summer Internship Program 2021, Gurugram Police / Cyber Crime Cell, July 2021